

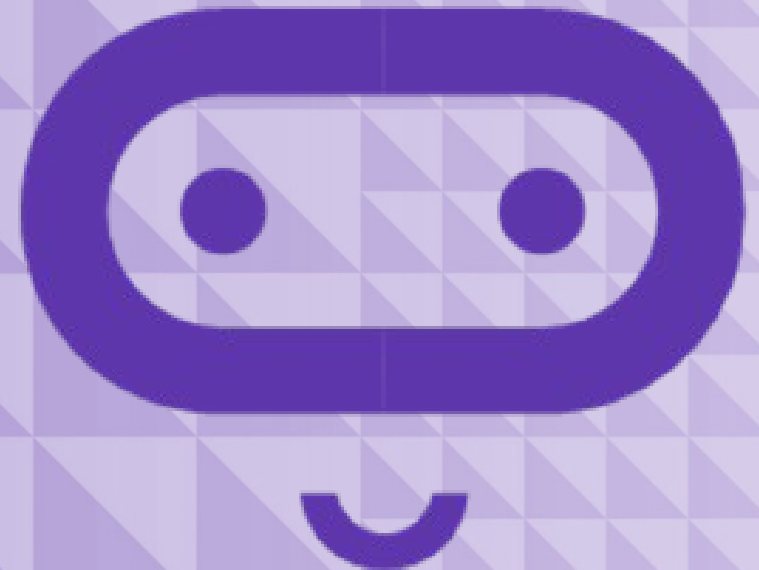
micro:bit

Introduction to Computer Science

Unit 3: Variables

Lesson A

Variables in daily life



What we'll learn in Lesson A

- The difference between variables and constants
- Five types of variables

Constants and variables

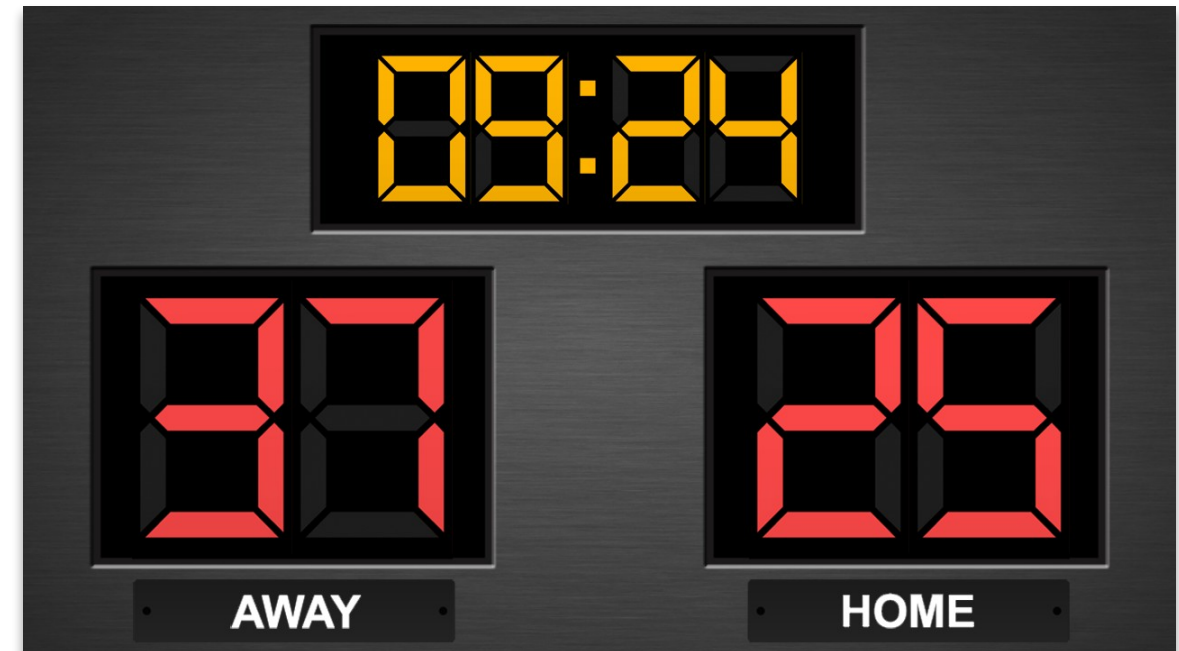
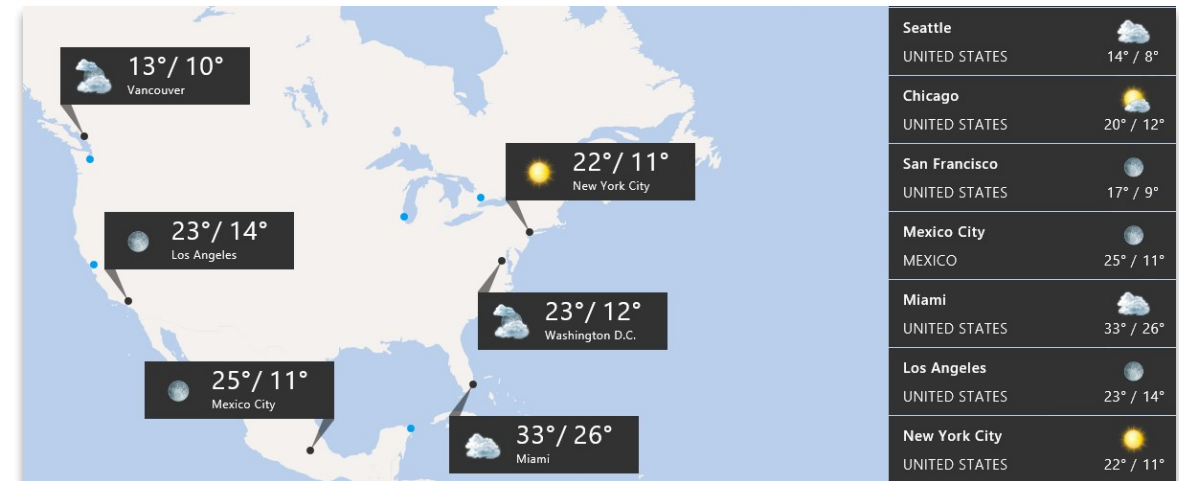
Constant

Information with **values that don't change** during a program

Variable

Information with **values that may change** during a program

- The type of variable is **assigned the first time you use a variable** depending on the kind of information it holds.
- From then on, you can only **change** the variable value to **another value of the same type**.



Types of variables

Number

Holds **numeric** data

String

Holds a string of **alphanumeric** characters

Boolean

Has only two possible values: **True or False**

Sprite

A special variable that holds two values for **the row and column (x, y)** to represent the location of a dot on the micro:bit screen

Array

A special variable that holds a **list** of multiple items

on start

set NumberVariable ▼ to 29

set StringVariable ▼ to "Hello!"

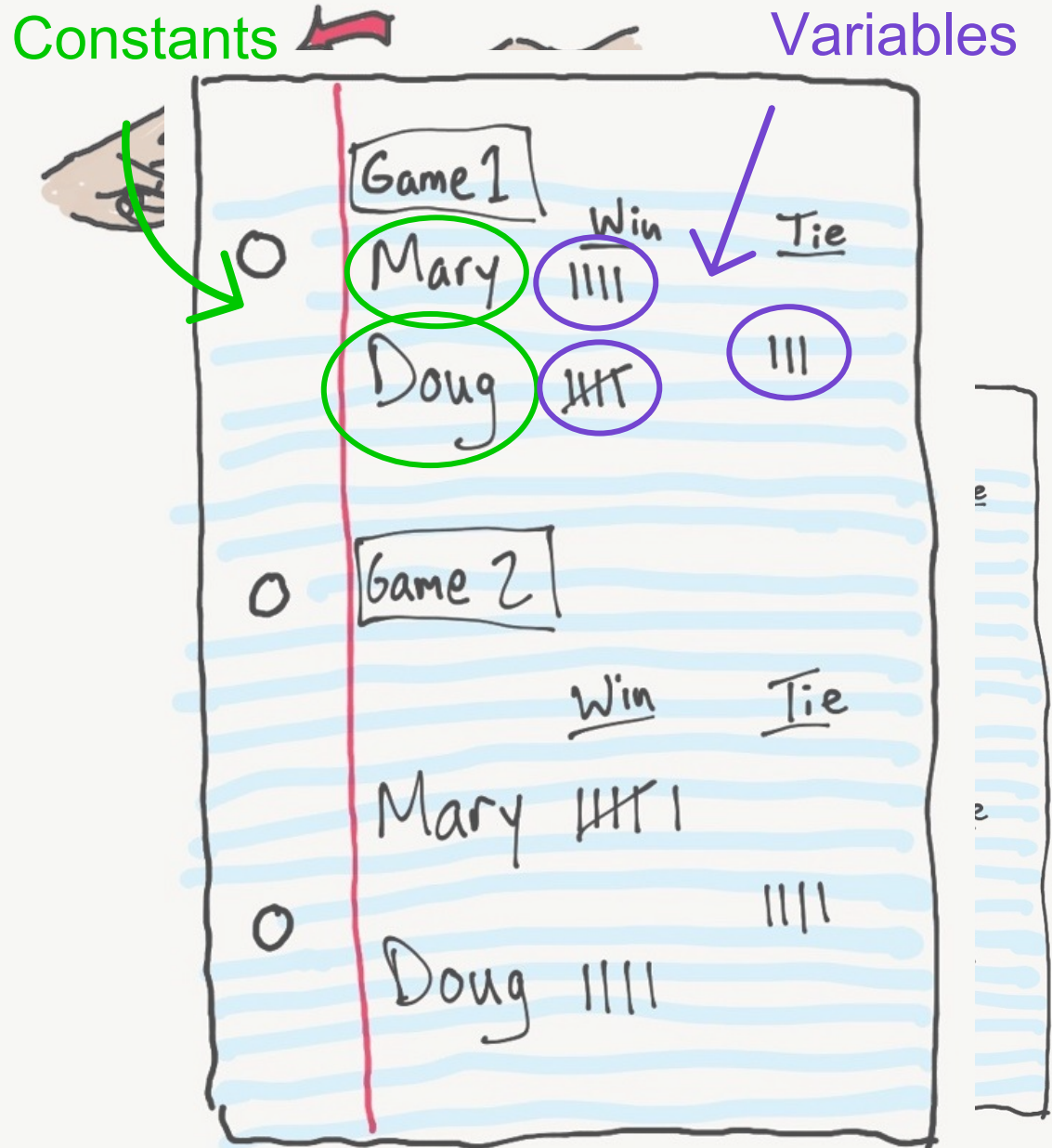
set BooleanVariable ▼ to true ▼

set SpriteVariable ▼ to create sprite at x: 2 y: 2

set List ▼ to array of "red" "orange" "yellow" - +

Understanding variables

- Play “Rock, Paper, Scissors” with a Partner.
- Keep score on a piece of paper.
- Identify the variables and constants on your scoresheet.



What we learned in Lesson A



- The difference between constants and variables in real life and in programming
- Five types of variables when coding the micro:bit

Next time: Lesson B

- Code the micro:bit to be a game scorekeeper
- Assess our learning with a quiz



Lesson B

Make a game scorekeeper



What we'll learn in Lesson B

- Code with string and number variables
- Assign meaningful names to variables
- Use math operations with variables
- How much we've already learned

Code a scorekeeper using variables

1. Create and name variables
2. Initialize the variable value
3. Update the variable value
4. Add user feedback
5. Show the final values
6. Add on with math operations



A hand-drawn scorecard on lined paper, divided into two sections by a vertical red line. Each section is titled 'Game 1' and 'Game 2' in boxes. The columns are labeled 'Win' and 'Tie' with underlines. Scores are represented by vertical bars (I for 1, II for 2, III for 3, IIII for 5). In Game 1, Mary has 4 wins and Doug has 3 wins and 1 tie. In Game 2, Mary has 4 wins and 1 tie, and Doug has 4 ties.

	<u>Win</u>	<u>Tie</u>
Game 1		
○ Mary	IIII	
Doug	III	III
Game 2		
Mary	IIII	I
○ Doug	IIII	IIII

Step 1. Create 3 variables

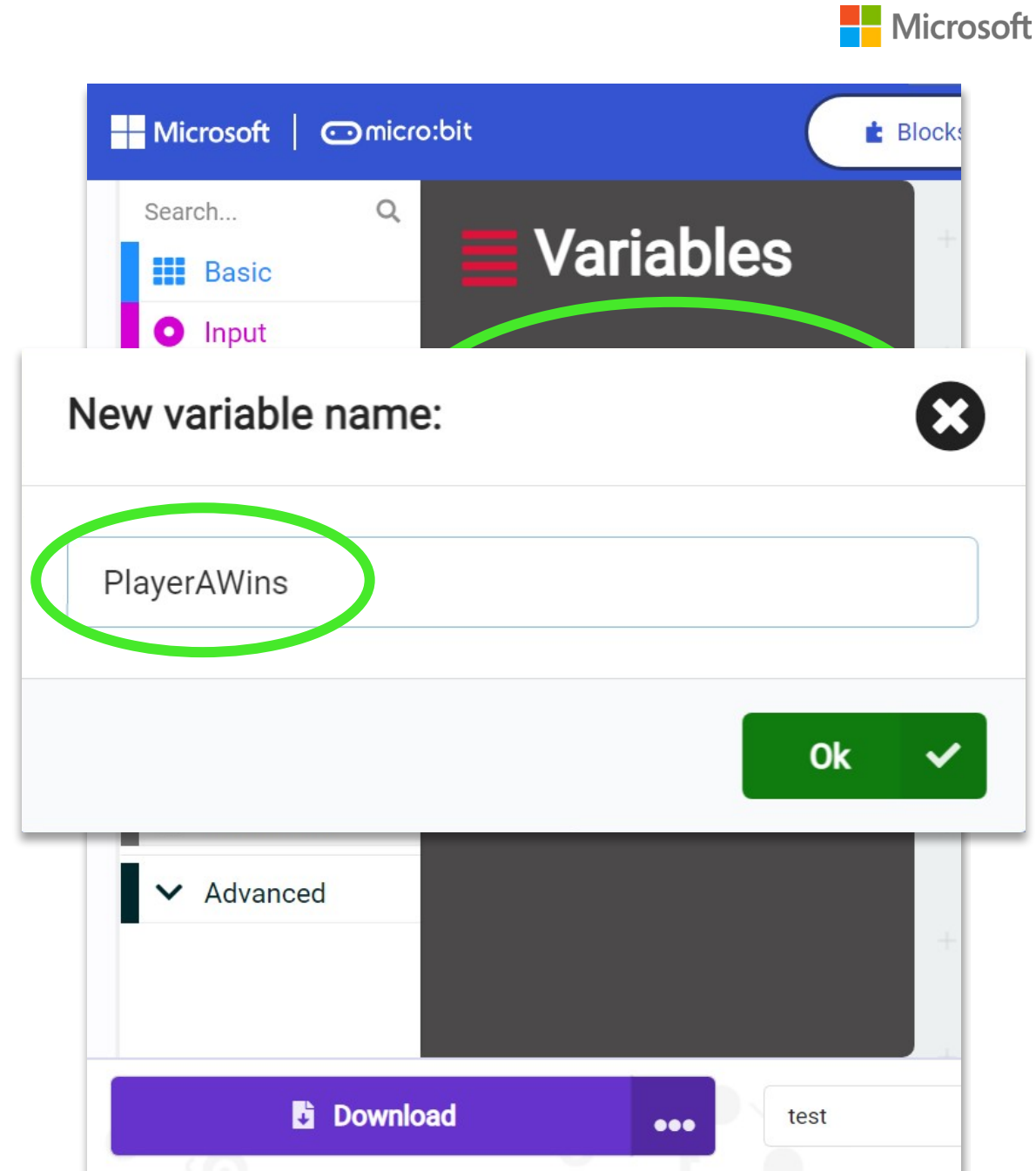
a. From the **Variables Toolbox** drawer, select the **Make a Variable** button.

b. Name your variables:

i. **PlayerAWins**

ii. **PlayerBWins**

iii. **PlayersTie**



Step 2. Initialize the variables

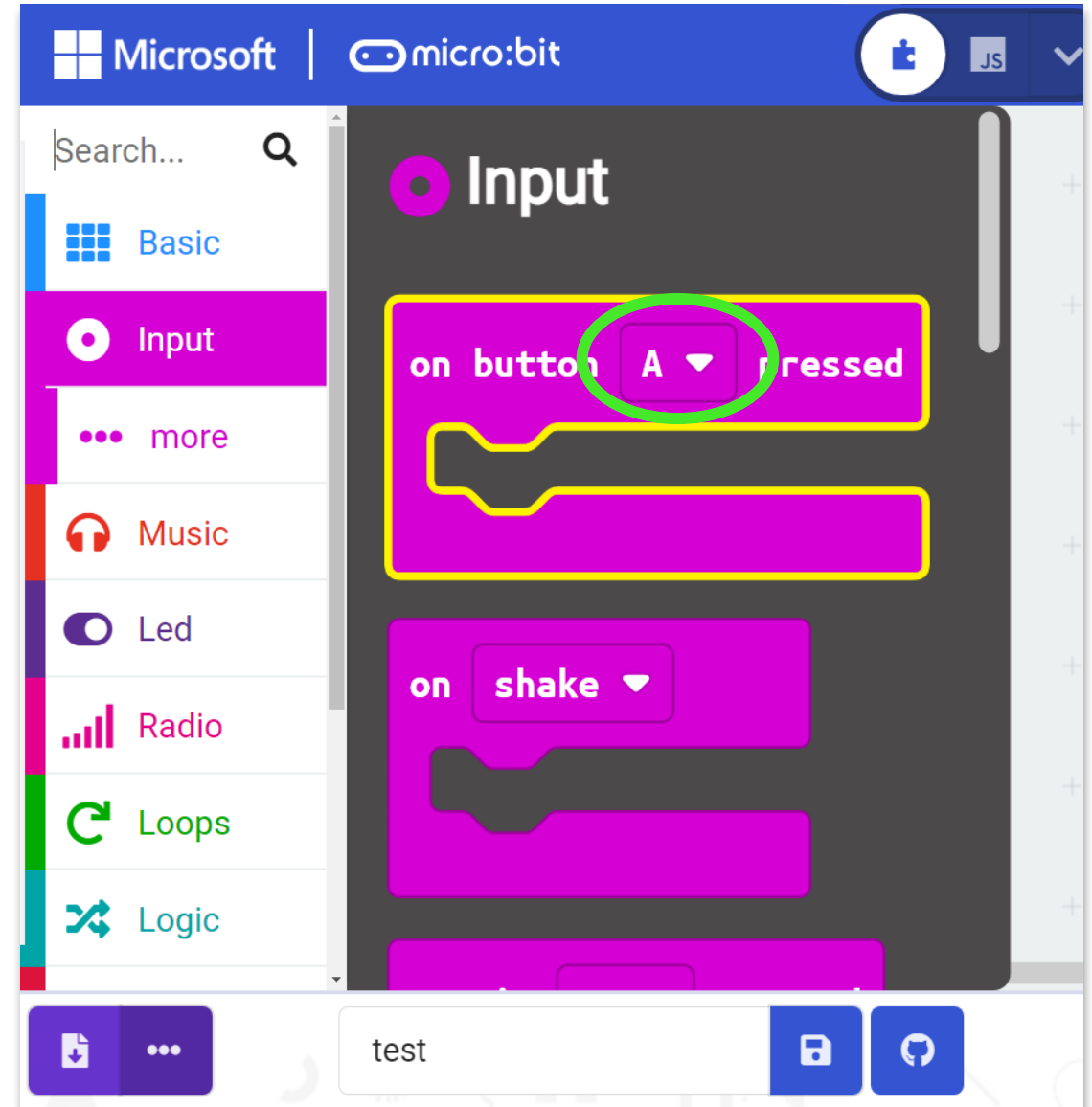
- Give your variables **starting values of 0**.
- From the **Variables Toolbox** drawer, drag out **3 Set variable blocks** onto the Workspace and drop them into the **On Start block**.
- Using the dropdown menus, **select your 3 variables**.



Step 3. Add to score

To add to the score when you press a button:

- From the **Input Toolbox** drawer, drag out **3 On Button Pressed** blocks onto the Workspace.
- Using the dropdown menus, select **Button B** and **Button A+B**.



Change variable values

Step 4. From the **Variable Toolbox** drawer, drag out 3 **Change Variable blocks** onto the Workspace.

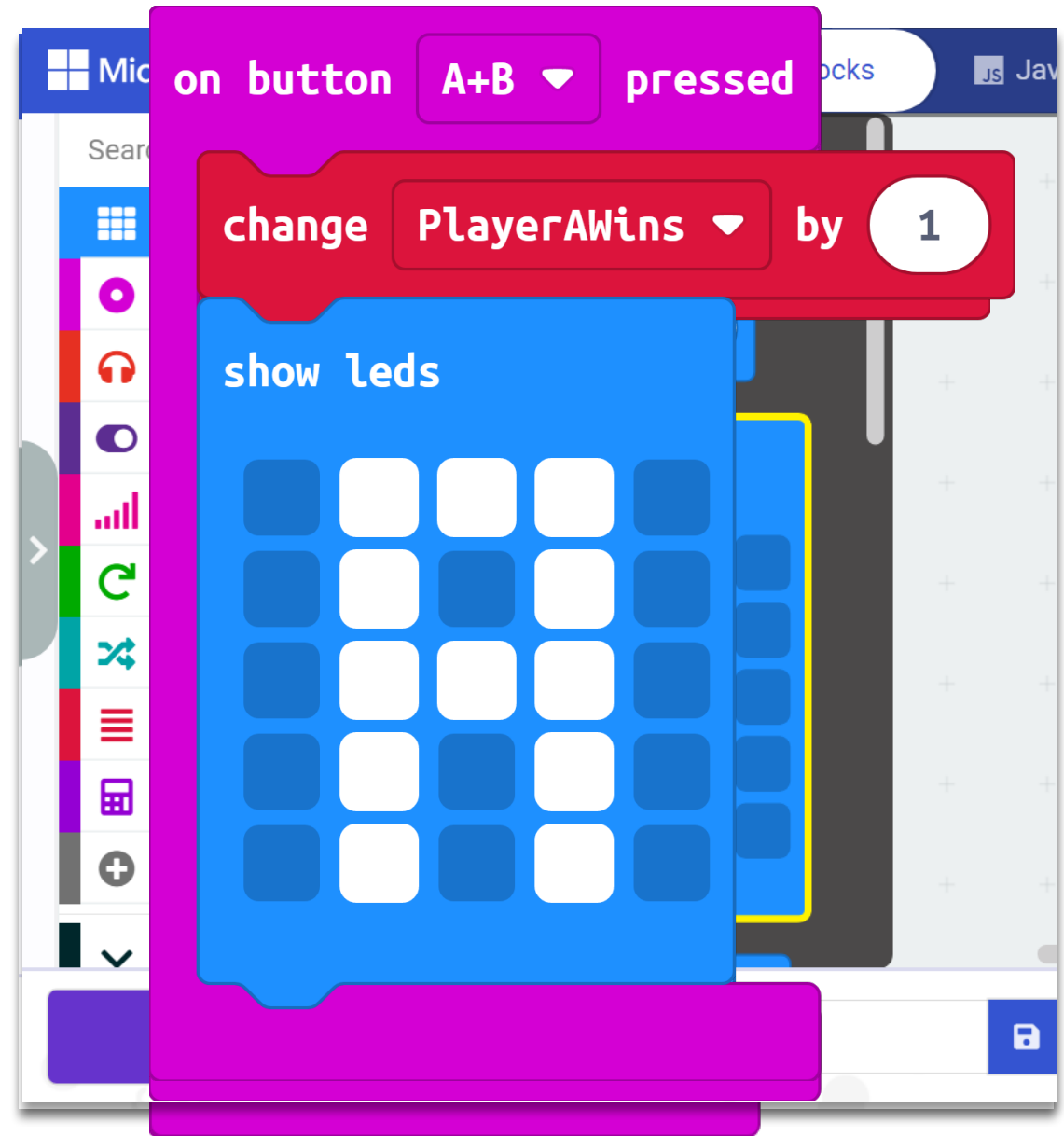
Step 5. Drop into each one into an **On Button Pressed** block.

Step 6. Using the dropdown menus, select each of the 3 **variable names**.



Step 7. Show which button changes what

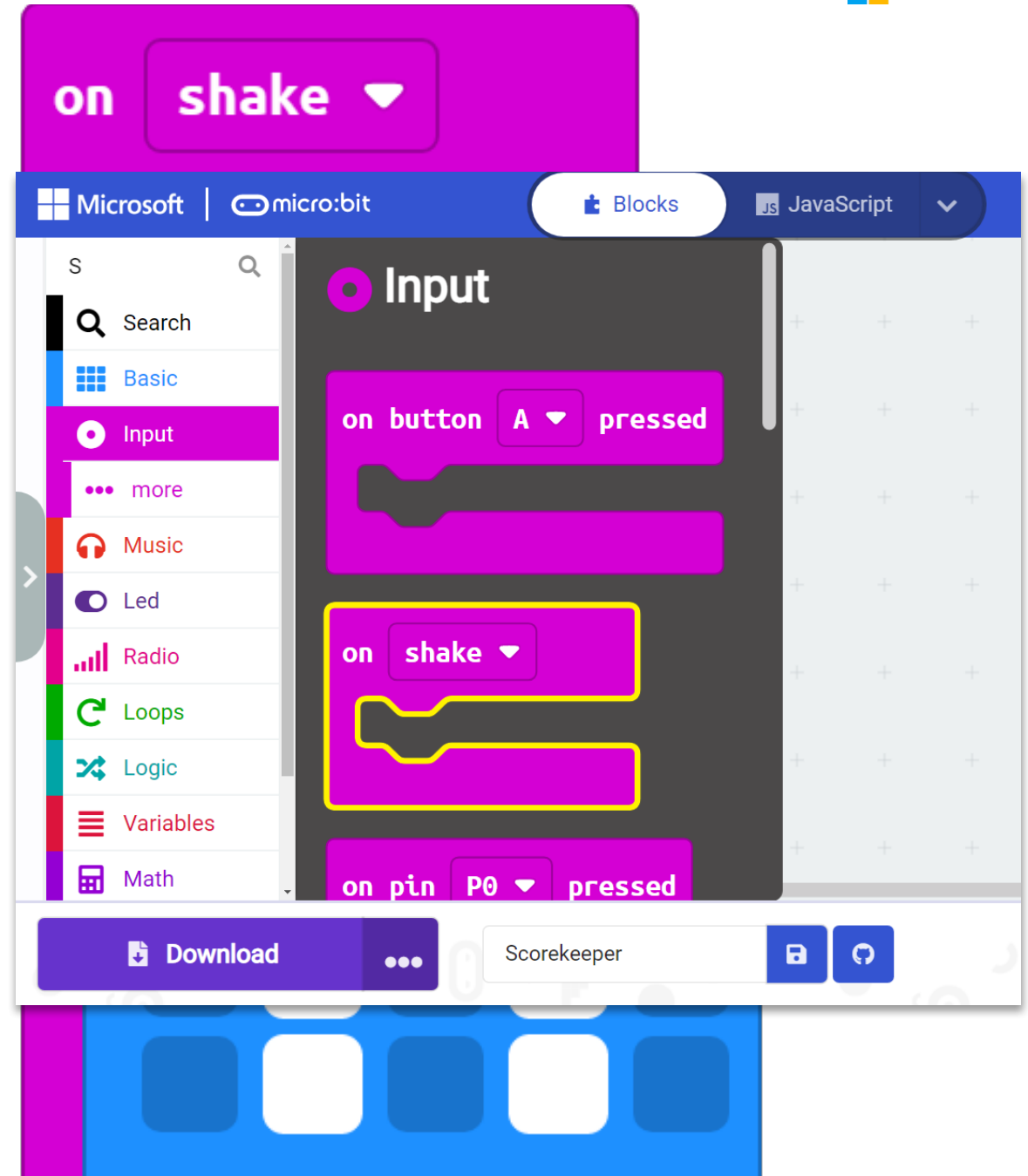
- From the **Basic Toolbox** drawer, drag out **3 Show LEDs blocks** onto the Workspace and drop them into each of the **On Button Pressed blocks**.
- Draw an **A**, **B** and **T** on the screen.
- From the **Basic Toolbox** drawer, drag **3 Clear Screen blocks** and drop them after the **Show LEDs blocks**.



Display the final values of each variable

Step 8. From the **Input Toolbox** drawer, drag out an **On Shake** block onto the Workspace.

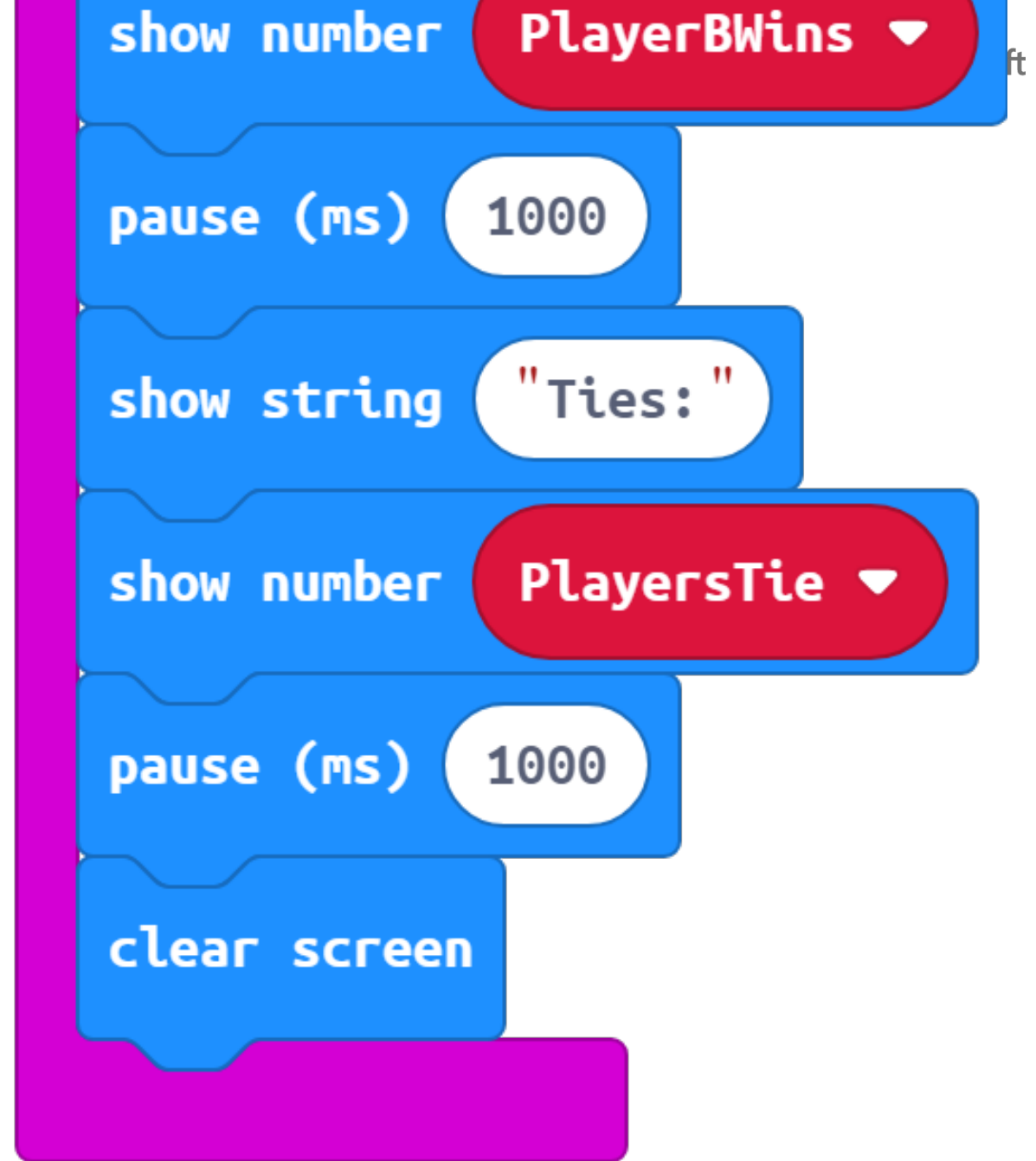
Steps 9-14. Use the **Basic blocks** and your **Variable blocks** to display the values.



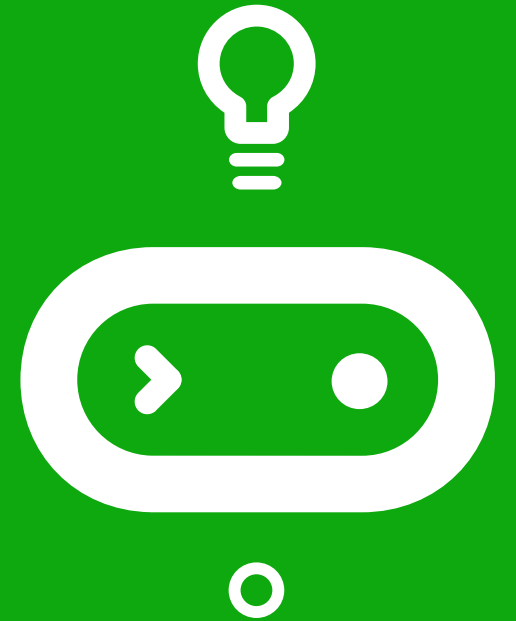
Display the final values of each variable

Step 8. From the **Input Toolbox** drawer, drag out an **On Shake** block onto the Workspace

Steps 9-14. Use the **Basic blocks** and your **Variable blocks** to display the values



Time for a quiz



What we learned in Lesson B

- Coded a game scorekeeper with string and number variables and math operations
- The importance of assigning meaningful names to variables



Next time: Lesson C

- Create and code a micro:bit program that counts something



Lesson C

Everything counts



What we'll learn in Lesson C

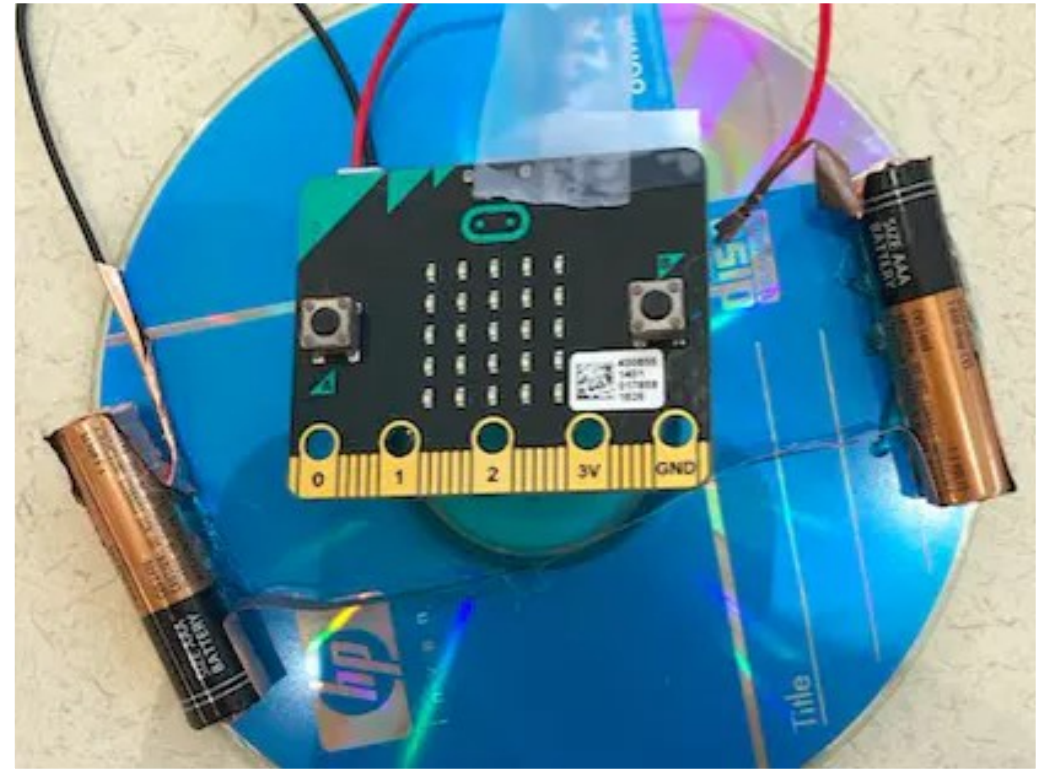
- Code a micro:bit program that counts something
- Design and build a gadget that uses the micro:bit

Project expectations

Think of a gadget that counts something.

Create a program to:

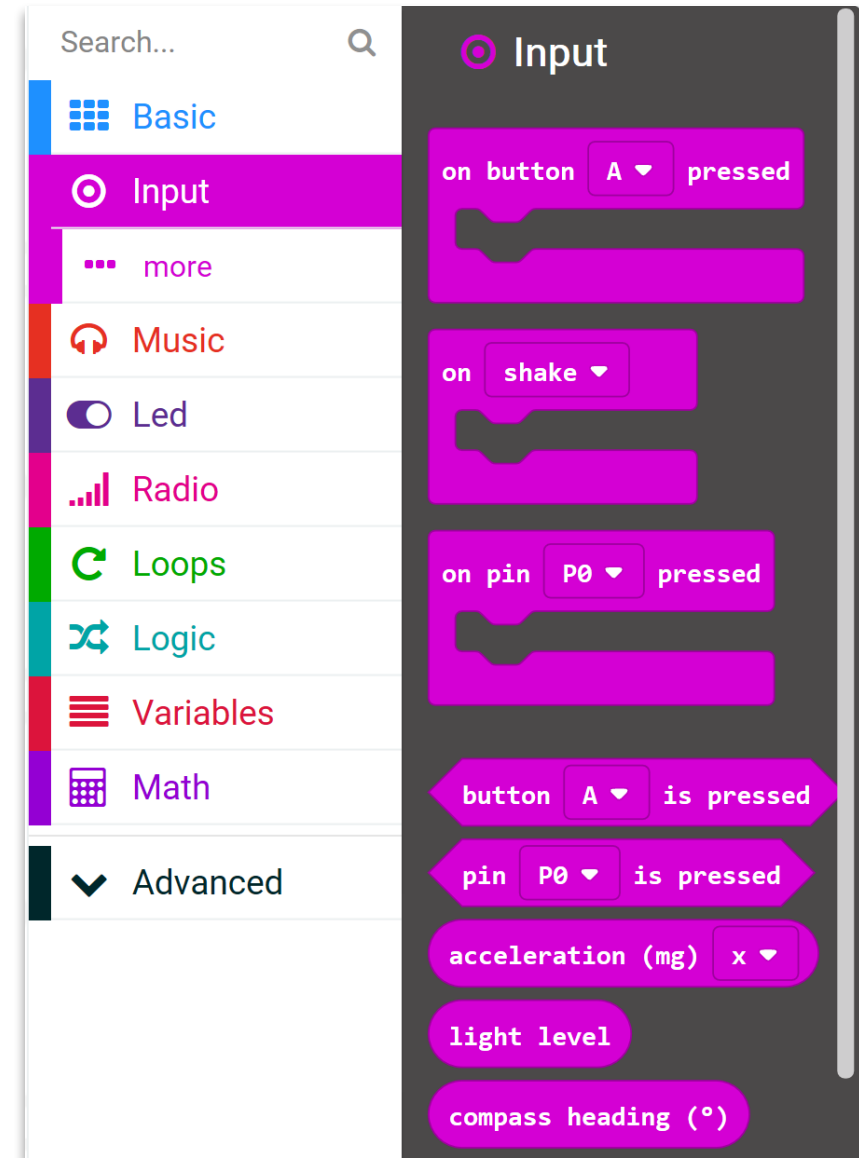
- Keep track of **inputs** with **at least 3 variables**
- Use **unique variable names** that clearly **describe the information values** the variables hold
- Provide **output** in some **visual and useful way**
- Use **math operations** to add, subtract, multiply, and/or divide at least two variables to give a useful output



Any guesses what this gadget is?

Available inputs

- Acceleration
- Light level
- Button is pressed
- Compass heading
- Temperature
- Running time
- On shake
- On button pressed
- On logo down
- On logo up
- One pin pressed
- On screen down
- On screen up
- Pin is pressed



Project ideas

🔗 Duct tape wallet

- Make a wallet or purse.
- Create a place for the micro:bit to fit securely.
- Use Button A to add dollars and Button B to subtract dollars. (Hint: The [duct tape wallet link](#) provides instructions!)

Extra mods

- Use other inputs to handle cents.
- Provide a way to display how much money is in the wallet in dollars and cents.



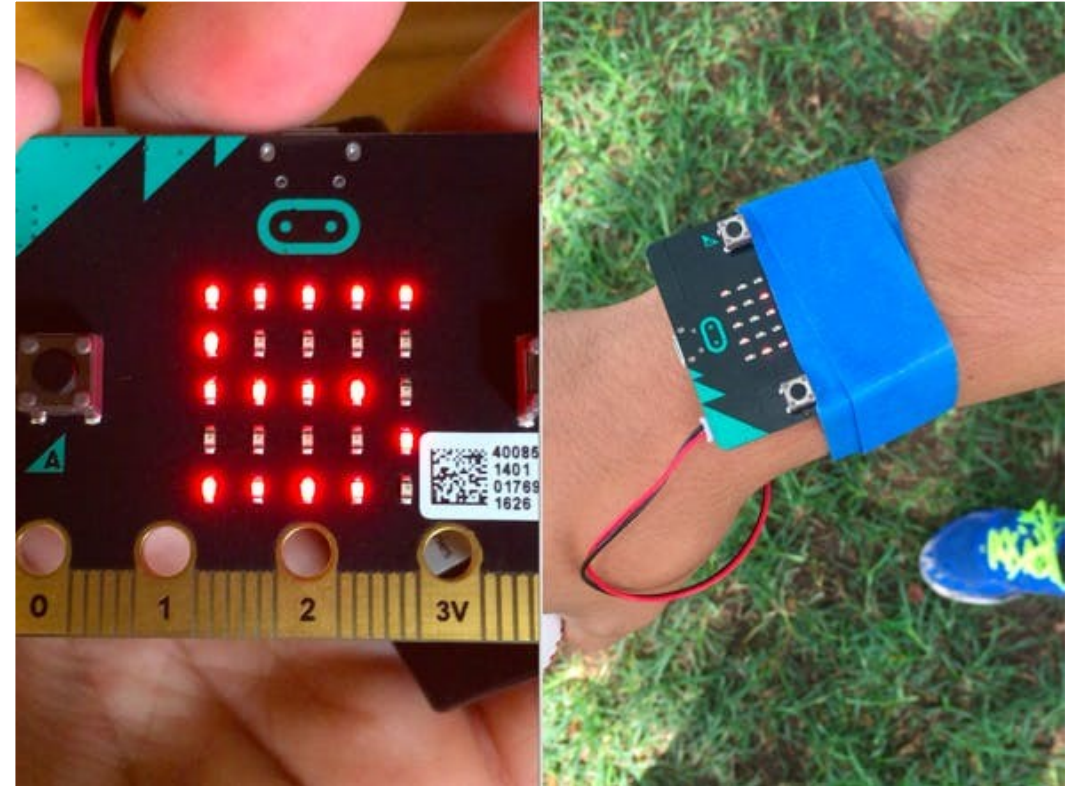
Another project idea

Umpire's baseball counter

- Make a wrist gadget to track how many pitches have been thrown to each batter.
- Use Button A to record the number of balls (up to four).
- Use Button B to record the number of strikes (up to three).

Extra mods

- Create a way to reset both variables to zero.
- Create a way to see the number of balls and strikes on the screen at the same time.



More project ideas

Shake counter

- Use the **on shake block** to detect when the micro:bit has been shaken and increment the variable accordingly.
- Attach the micro:bit to a lacrosse stick or track how many times you have successfully thrown a ball in the air and caught it.

Extra mods

- Have the micro:bit make a sound of increasing pitch every time you successfully catch the ball.

Even more project ideas

Pedometer

- See if you can count your steps while running or doing other physical activities carrying the micro:bit.
- Decide on the best place to mount the micro:bit.

Extra mods

- Design a wearable band or holder that can carry the micro:bit securely so it doesn't slip out during exercise.

And even more project ideas

Calculator

- Create an adding machine.
- Use Button A to increment the first number, and Button B to increment the second number.
- Use shake or Buttons A + B to add the two numbers and display their sum

Extra mods

- Find a way to select and perform other math operations

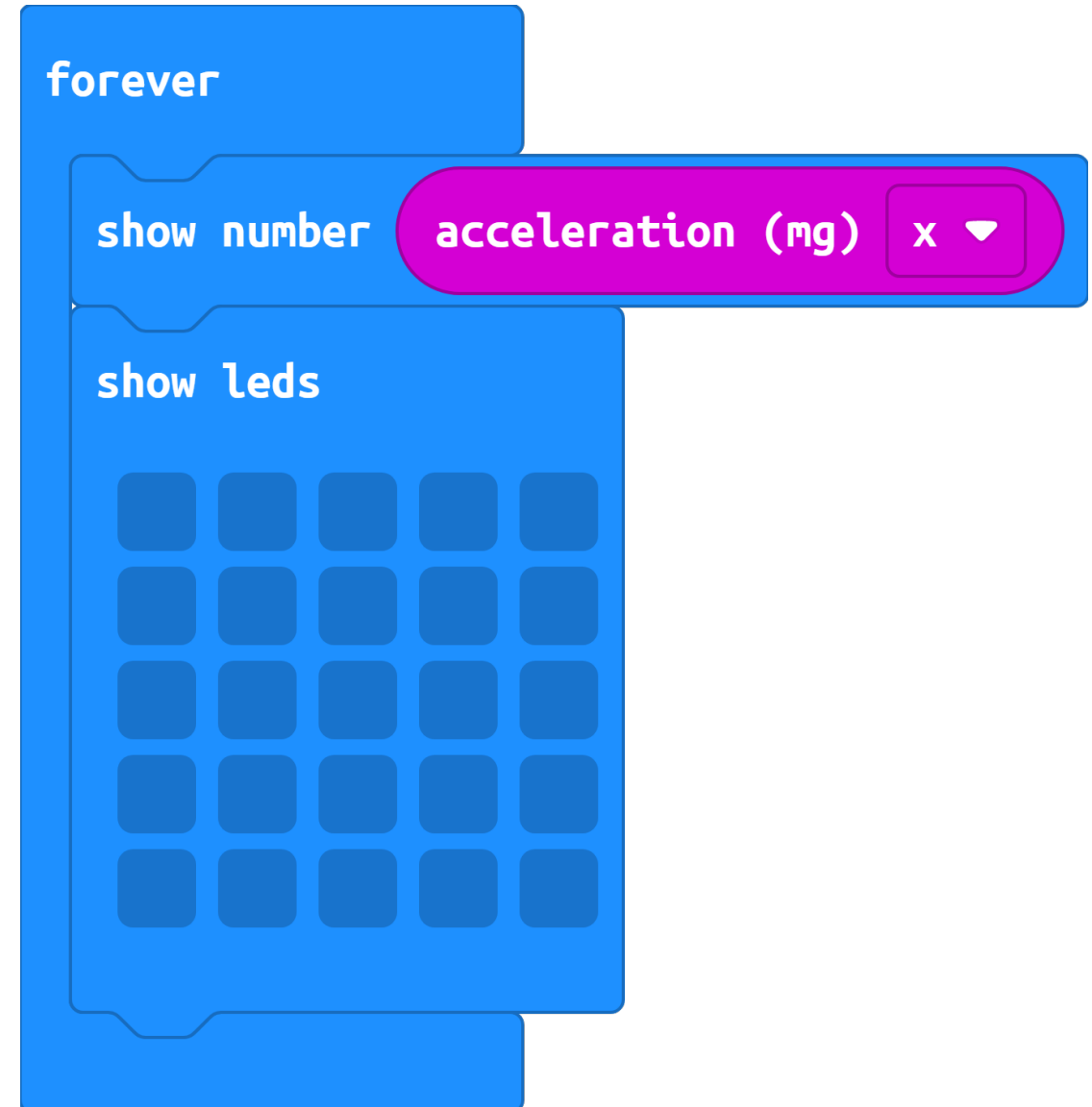
The process

Understand the problem

- Get clear on the purpose
- Prototype
- Start brainstorming
- Sketch a variety of different ideas
- Far out, impractical ideas are good

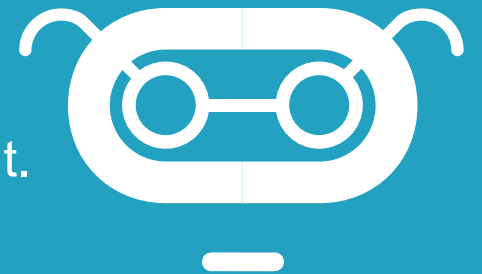
Code the program

- Use the simulator to do your programming
- Test out a variety of different ideas



Reflection Diary

- What was the problem you were trying to solve with this project?
- What were the variables that you used?
- What mathematical operations did you perform on your variables? What information did you provide?
- Describe what the physical component of your micro:bit project was (or provide a pic).
- How well did your prototype work? What did you like? What would you change?
- What was surprising to you about the process of creating this project?
- Describe a difficult point while designing this project and explain how you resolved it.
- Publish your MakeCode project and include the link.



What we learned in Lesson C

- Coded a micro:bit program that counts something
- Designed and built a gadget that uses the micro:bit

